

Ali Omrani

Current Position

Senior Machine Learning Engineer & Tech Lead
Snap Inc., Los Angeles, CA

Contact Info

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RESEARCH INTERESTS

- LLM & Agentic AI Evaluation and Safety in Production Deployment
- Socially-Aware NLP: Bias Mitigation, Harm Detection, Business Integrity
- Agentic AI Systems: Robustness, Failure Detection, and Environment Feedback Loops
- Human-in-the-Loop Machine Learning and Active Learning

EDUCATION

Ph.D. in Computer Science

Jan. 2020 – May 2024

University of Southern California

Thesis: *Countering Problematic Content in Digital Space*

Committee: Morteza Dehghani, Xiang Ren, Robin Jia, Payam Piray, Jeffrey Sorensen (Google)

M.Sc. in Computer Science

Jan. 2020 – Jan. 2022

University of Southern California

Courses: Data-centric NLP, Robustness in NLP, Applied NLP, Representation Learning, Theory of Machine Learning, Econometrics, Multilevel Modeling, Advanced Algorithms

B.Sc. in Computer Engineering

Sept. 2014 – Sept. 2019

Department of Computer Engineering, University of Tehran

PI: Majid Nili Ahmadabadi

PUBLICATIONS

- [1] Osborne, M.R., Abdurahman, S., **Omrani, A.**, Trager, J.P., Dehghani, M. (2026). [Two wrongs is what makes it more right: How retaliatory incivility receives social leniency](#) — *Journal of Experimental Social Psychology*, 125, 104913.
- [2] Trager, J., Ziabari, A.S., Davani, A.M., Golazizian, P., Karimi-Malekabadi, F., **Omrani, A.**, ... , Dehghani, M. [The Moral Foundations Reddit Corpus](#) — Proceedings of the 15th Conference on Language Resources and Evaluation (LREC 2026).
- [3] Abdurahman, S., Atari, M., Karimi-Malekabadi, F., Xue, M.J., Trager, J., Park, P.S., Golazizian, P., **Omrani, A.**, Dehghani, M. [Perils and Opportunities in Using Large Language Models in Psychological Research](#) — *PNAS Nexus*.
- [4] **Omrani, A.***, Ziabari, A.*, Golazizian, P., Sorensen, J., Dehghani, M. [Towards a Unified Framework for Adaptable Problematic Content Detection via Continual Learning](#) — NAACL 2024, Workshop on Online Abuse and Harms (WOAH).
- [5] Ziabari, A., **Omrani, A.**, Golazizian, P., Hejabi, P., Kennedy, B., Dehghani, M. (2024). [Reinforced Multiple Instance Selection for Speaker Attribute Prediction](#) — NAACL 2024.
- [6] Golazizian, P.*, Ziabari, A.S., **Omrani, A.**, Dehghani, M. [Cost-Efficient Subjective Task Annotation and Modeling through Few-Shot Adaptation](#) — Findings of EMNLP 2024.
- [7] **Omrani, A.**, Ziabari, A., Yu, C., Golazizian, P., Kennedy, B., Atari, M., Ji, H., Dehghani, M. (2023). [Social-Group-Agnostic Bias Mitigation via the Stereotype Content Model](#) — ACL 2023. [Nominated for Best Paper Award \(top 1.6% of submissions\)](#).
- [8] Osborne, M., **Omrani, A.**, Dehghani, M. (2023). The sins of the parents are to be laid upon the children: biased humans, biased data, biased models — *Perspectives on Psychological Science*.
- [9] Atari, M., Mehl, M.R., Graham, J., Doris, J.M., Schwarz, N., Davani, A.M., **Omrani, A.**, Kennedy, B., et al. (2023). [The paucity of morality in everyday talk](#) — *Scientific Reports*.
- [10] Mostafazadeh Davani, A., **Omrani, A.**, Kennedy, B., Atari, M., Ren, X., Dehghani, M. (2021). [Improving Counterfactual Generation for Fair Hate Speech Detection](#) — ACL 2021, Workshop on Online Abuse and Harms (WOAH).

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- [11] Kennedy, B., Atari, M., Mostafazadeh Davani, A., Hoover, J., **Omrani, A.**, Graham, J., Dehghani, M. (2021). [Moral Concerns are Differentially Observable in Language](#) — *Cognition*.
- [12] Kennedy, B., Atari, M., Mostafazadeh Davani, A., Yeh, L., **Omrani, A.**, . . . , Dehghani, M. (2020). [Introducing the Gab Hate Corpus: Defining and applying hate-based rhetoric to social media posts at scale](#) — *Language Resources and Evaluation*.

UNDER REVIEW & IN PREPARATION

- [1] **Omrani, A.**, Lince, A., Akkawi, A., Mahmoody, A. Human-in-the-Loop Fraud Trend Detection with LLMs — Under Review at KDD.
- [2] Karimi-Malekabadi, F., Oyserman, D., Ziabari, A.S., **Omrani, A.** [What Doesn't Kill Us: Harsh Ecologies Beget Difficulty-as-Improvement Norms](#) — R&R at *Nature Human Behavior*.
- [3] Abdurahman, S., **Omrani, A.**, Hoover, J., Dehghani, M., Jordan, J., Kteily, N. Twitter users preferentially reinforce disproportionate condemnation of MeToo transgressions, but privately prefer proportionality — Under Review at PNAS
- [4] Atari, M., **Omrani, A.**, Dehghani, M. [Contextualized Construct Representation: Leveraging Psychometric Scales to Advance Theory-Driven Text Analysis](#) — In Preparation.
- [5] **Omrani, A.**, Sorensen, J., Dehghani, M. [Parameter Efficient Learning with Noisy Labels](#) — Under Review.
- [6] **Omrani, A.**, Karimi-Malekabadi, F., Rahmati, E., Ziabari, A., Sorensen, J., Dehghani, M. [Cultural Diversity Enhances Offensive Language Detection in Multilingual Models](#) — Under Review.

HONORS AND AWARDS

- Agentic AI Hackathon Winner (Snap) 2025
- Best Poster Award, Society for Personality and Social Psychology (SPSP) 2023
- Best Bachelors Thesis Award, [Tap30](#) and University of Tehran 2019
- Ranked **2nd** in Computer Engineering Students, University of Tehran 2018
- DAAD Scholarship for Summer Internship at [Fraunhofer IDMT](#) 2017
- Faculty of Engineering Award, University of Tehran 2016
- Ranked **115th** among 200,000+ participants in nationwide university entrance exam 2014
- **3rd** Place in RoboCup Iran Open 2012 Junior Soccer League 2012
- Member of the National Organization for Development of Exceptional Talents (NODET) 2007

WORK & RESEARCH EXPERIENCE

Senior Machine Learning Engineer & Tech Lead Feb. 2026 – Present
Advertiser Understanding & Personalization, Snap Inc., United States

Tech Lead – Advertiser Understanding and Personalization

- Spearheaded [agentic advertiser profiling](#), segmentation, and similarity search systems using semantic IDs and hybrid search, powering personalized onboarding and campaign creation at scale; the platform serves as core infrastructure consumed by recommendations, fraud detection, agentic ad review, and sales insights.
- Defined optimality in recommendations and delivered multiple ML and agentic launches driving double-digit revenue lift, significant gains in advertiser retention, and measurable improvements in campaign setup efficiency.
- Own end-to-end roadmaps spanning advertiser profiling, campaign recommendations, and optimality — from problem definition through launch and iteration; drive org-level AI-driven recommendation strategy, aligning with Product, Marketing Science, Data Science, and Engineering on evaluation frameworks, phased rollouts, and production safety standards.

Machine Learning Engineer Jun. 2024 – Jan. 2026
Advertiser Experience ML, Snap Inc., United States

Ads Understanding

- Developed and scaled a multimodal AI system that converts ad images, video, and audio into structured text descriptions using large multimodal models, achieving near-100% ad coverage

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across Snap’s full ads inventory.

- System serves as the backbone for downstream applications including fraud detection, creative discovery, and content classification — a foundational infrastructure layer consumed across multiple product teams.

LLM Evaluation Infrastructure

- Architected shared LLM evaluation infrastructure adopted across the team: a shadow launch framework for safe GenAI evaluation on live production traffic, LLM observability standards, and custom model evaluation pipelines.
- Designed active learning loops leveraging internal human annotators for continuous model improvement, enabling data-efficient post-deployment refinement without full retraining cycles.

GenAI Content Moderation

- Conducted R&D on post-training multimodal LLMs for automated detection of policy-violating ad content, significantly increasing tagging automation and reducing operational review costs.
- Designed evaluation protocols and human annotation pipelines to validate model behavior on edge cases and distribution shifts.

Fraud Trend Detection

- Designed a novel human-in-the-loop AI pipeline for real-time fraud trend detection, combining LLM-based signal extraction with structured human feedback loops to surface emerging fraud patterns at scale.
- Authored a research paper on the system; currently under review at KDD.

Ad Relevance

- Built an LLM-based ad relevance scoring system demonstrating significantly stronger correlation with true user relevance compared to prior proxy signals; winner of internal GenAI Hackathon.

Mentorship & Knowledge Sharing

- Mentored junior engineers and summer interns; guided an apprentice to full-time conversion.
- Delivered org-wide presentations on LLMs and agentic systems; led knowledge-sharing sessions on productionizing GenAI at scale.

Software Engineering Intern

May 2023 – Aug. 2023

Identity Graph, Snap Inc., United States

Manager: Shipeng Gong Mentors: Bahador Yeganeh & Houshmand Shirani-Mehr

Designed, developed, and deployed a language model to learn user journey patterns on advertiser websites. Integrated this model into the identity framework, achieving a 3% boost in precision; adopted in production.

Software Engineering Intern

May 2022 – Aug. 2022

Business Integrity, Snap Inc., United States

Manager: Xiaojia Zhao Mentor: Liquan Yu

Constructed a machine learning pipeline to analyze and categorize advertisement website content based on a comprehensive tag taxonomy. Attained $F1 > 0.8$ for most tags; deployed in production for tag suggestions, streamlining annotation and enhancing recall.

Research Assistant

Jan. 2020 – May 2024

Morality and Language Lab (MOLA), USC, United States

Designed, developed, and validated NLP systems at the intersection of machine learning and social science. Published work spans bias mitigation, hate speech detection, continual learning for content moderation, and LLM evaluation for subjective annotation tasks. Work published at ACL, NAACL, EMNLP, *Cognition*, and *Scientific Reports*.

Research Assistant

Aug. 2018 – Aug. 2019

Cognitive Systems Lab, University of Tehran, Iran

Worked on attention mechanisms in medical decision support systems under the supervision of Majid Nili Ahmadabadi. Thesis demonstrated improvements in interpretability and generalization; received Best Thesis Award from University of Tehran and Tap30.

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Research Intern

Jun. 2017 – Sept. 2017

Music Information Retrieval Group, Fraunhofer IDMT, Germany

Designed and developed a singing voice detection system from weak labels using deep convolutional neural networks on audio spectrograms. Applied post-hoc analysis methods to improve performance and reduce model size (DAAD-funded).

Research Intern

Jun. 2016 – Sept. 2016

Data Science Lab, University of Tehran, Iran

Applied graph analysis to Call Data Records (CDR) of Iran's largest mobile operators under the supervision of Behnam Bahrak. Detected anomalies and linked them to real-world events including popular sports broadcasts, spammer activity, and public holidays.

AD-HOC REVIEW

- Association for Computational Linguistics (ACL)
- Empirical Methods in Natural Language Processing (EMNLP)
- Cognitive Science Society (CogSci)

TEACHING EXPERIENCE

- Tutorial – **Theory-Driven Text Analysis for Social Psychologists** SPSP 2023
- Guest Lecture – **Applied Machine Learning** USC 2022
- Teaching Assistant – **Analysis of Algorithms** Fall 2022, Summer 2020, Spring 2020
- Head Teaching Assistant – **Artificial Intelligence** Spring 2018
- Head Teaching Assistant – **Software Testing** Fall 2018
- Teaching Assistant – **Network Security** Spring 2018, Fall 2018
- Teaching Assistant – **Operating Systems** Spring 2018, Fall 2018, Spring 2017
- Teaching Assistant – **Databases** Spring 2018, Fall 2018, Spring 2017
- Teaching Assistant – **Engineering Probability and Statistics** Fall 2016
- Teaching Assistant – **Introduction to Computing and Programming** Fall 2015

SKILLS

- **Programming:** Python, C, C++, Java, R, Matlab
- **Machine Learning:** PyTorch, TensorFlow, Transformers (HuggingFace), Scikit-learn, Pandas, NumPy
- **LLMs & Agentic Systems:** LLM fine-tuning, RLHF/post-training, RAG, agentic pipelines, multimodal models, LLM & Agentic evaluation, shadow launch frameworks, active learning
- **Search & Retrieval:** Semantic embeddings, hybrid search, vector databases, semantic ID systems
- **Cloud & Infrastructure:** BigQuery, Airflow, Vertex AI, Google Cloud Storage (GCS)
- **Statistics:** Multi-level modeling, econometrics, Bayesian methods
- **Languages:** English (fluent), Persian (native)